

A Macroeconomic Approach to Optimal Fertility Across Nations

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Abstract

Should countries aim for the replacement level of fertility? Or are optimal fertility rates close to the replacement level? Does the answer differ across countries, and if so, why? This paper explores these questions through the lens of a dynastic model that captures multiple channels through which fertility affects long-run national income and welfare. I calibrate and estimate the model parameters using data from nine high-income countries. Solving the planner's problem for each country, I find that although the optimal fertility rates exceed the current equilibrium rates, they still remain below the replacement level in many countries. This is particularly true for countries such as Germany, Japan, and Spain, where higher child penalties raise the social costs of increased fertility through the associated output losses.

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1 Introduction

In many countries, total fertility rates are far below the replacement level—the fertility rate required to keep the population size stable over time—which is approximately 2.1 in high-income countries. Sustained low fertility leads to population aging and raises serious macroeconomic and fiscal concerns, including the sustainability of social security systems, declines in the labor force, slower economic growth, consequently lower living standards. In response, many governments have introduced or expanded pro-natal policies, such as childcare subsidies, parental leave, and birth subsidies, aiming for higher fertility.

Yet, we still lack a clear understanding of (1) whether current fertility is in fact “too low,” (2) what the target of pro-natal policies should be, including whether we should aim for the replacement rate or something else, and (3) the determinants of such an *optimal* fertility level—that is, how it is shaped by preferences, technology, and social institutions that differ across countries. This paper therefore asks the following questions: Should countries aim for the replacement level of fertility? Or are optimal fertility rates close to, above, or below the replacement level? Do the answers differ across countries, and if so, why? These questions are important because, without understanding these points, the economic rationale and the goal of pro-natal policies remain unclear.

My contribution is to address these questions through two methodological advances. First, I construct a macroeconomic model that captures multiple channels through which households’ fertility choices affect national income and welfare: positive externalities arising from a larger stock of ideas (Jones, 2022) and increased labor force, as well as negative externalities related to the quantity–quality trade-off and the child penalty, both of which reduce aggregate labor input and raise the equilibrium tax rate. Building on this baseline model, I further examine additional channels—such as pollution and climate change—as a robustness check. Second, I quantify the model using cross-country data to examine how optimal fertility differs across countries and what drives those differences.

Using the quantitative model, I find that although the optimal fertility rates—the solutions to the planner’s problem—exceed the equilibrium rates, they still remain far below the replacement level in many countries. This is particularly true for countries such as Germany, Japan, and Spain, where higher child penalties raise the social costs of increased fertility through the associated output losses. These results suggest that the justification for pro-natal policy is considerably weaker in the absence of measures that reduce the social costs of higher fertility, such as the child penalty. The key results are robust to extensions of the model that incorporate additional channels, such as pollution and climate change.

This study is most closely related to the literature examining efficiency in models with endogenous fertility choices (e.g., Yew and Zhang, 2009; Schoonbroodt and Tertilt, 2014; Córdoba and Liu, 2022; Jones, 2022; Gerlagh, Lupi, and Galeotti, 2023; Weil, 2023).¹ This literature highlights the existence of wedges and market failures through which equilibrium

¹This study follows the standard approach in the literature and adopts a dynastic model to analyze optimality in models with endogenous fertility. However, analyzing optimal or efficient allocations in such models presents a conceptual challenge because changes in fertility alter both the number and the identity of individuals whose welfare must be considered. For this reason, a separate strand of the literature develops efficiency criteria suited to environments with endogenous population (e.g., Golosov, Jones, and Tertilt, 2007; Conde-Ruiz, Giménez, and Pérez-Nievas, 2010; De la Croix and Doepke, 2021).

fertility can deviate from the socially optimal level, either being too high or too low. Building on these insights, my contribution is to incorporate multiple such channels into a unified framework for analyzing optimal fertility. These channels include (i) a positive externality whereby a larger population increases the stock of non-rival ideas and thus raises aggregate income (Jones, 2022), (ii) a long-run demographic effect, where a larger working-age population increases output per-capita and expands the tax base, and (iii) the quantity–quality trade-off in human capital investment (Yew and Zhang, 2009). I also confirm that the key results are robust to extensions of the baseline model that incorporates additional channels, such as pollution and climate change (Gerlagh, Lupi, and Galeotti, 2023) and fixed resources (Córdoba and Liu, 2022). Another contribution is to quantify these channels using cross-country data and examine how and why the optimal fertility differs across countries.

2 Model

I construct a dynastic model featuring three sectors—households, a representative firm, and the government. Section 2.1 outlines the model environment. Section 2.2 formulates the household problem in the decentralized economy and defines the associated equilibrium concept. Section 2.3 presents the planner’s problem and discusses the externalities that the planner internalizes.

2.1 Environment

Preference: This model builds on a dynastic framework in the tradition of the existing literature (e.g., Jones, 2022). Households derive utility from consumption (c), leisure (l), and having children (n'). They discount the utility of their children by a function $\Psi(n')$, which is increasing in the number of children and satisfies $\Psi(0) = 0$ and $\Psi(n') < 1$ for any $n' > 0$. Having children is costly in several ways. First, each child reduces parental working hours by a fraction δ , which I refer to as the child penalty. Parents also choose how much to invest in each child’s education (e), which increases the child’s human capital according to a human capital production function $q' = \Phi(e)$, thereby raising their future income. Parents make these decisions based on their disposable income, determined by the equilibrium wage rate w , hours worked, and human capital, with income taxed at rate τ .

Technology The representative firm produces final goods using labor N as the sole input, with the production function given by $Y = ZN$. The technology term $Z = n^\beta A$ represents the overall productivity level, where the population mass of parents n serves as a proxy for the equilibrium population size. This specification captures the idea that a larger population contributes to idea creation, with $\beta > 0$ governing the strength of this spillover, and A representing other technological factors. Under competitive markets, the equilibrium wage equals the marginal product of labor, $w = Z$. Since Z increases with population size, the wage is an increasing function of fertility. Consequently, fertility choices generate a positive externality through this channel, which households do not internalize. As a result, equilibrium fertility may be lower than the socially optimal level.

Government The government collects a labor income tax to finance public expenditure G . The tax rate τ is determined to satisfy the government's budget constraint, $\tau wN = G$, implying that the equilibrium tax rate is given by $\tau = G/(wN)$. In addition to the positive externalities arising from idea spillovers, fertility may generate further positive externalities because, ceteris paribus, it increases the labor force and thereby reduces the tax rate. However, fertility can also produce negative fiscal externalities, since having more children entails child penalties and reduces educational investment per child, both of which lower labor efficiency and raise the equilibrium tax rate. Whether the net fiscal externality of fertility is positive or negative is ultimately a quantitative question, the answer to which depends on several parameters governing these countervailing forces, such as the magnitude of the child penalty and the extent of the quality-quantity trade-off.

2.2 Decentralized Economy

In the decentralized economy, a mass of households choose consumption, leisure, the number of children, and education spending, subject to a budget constraint. The mass of households, n , and their human capital level, q , are predetermined by the decisions of their ascendants. The household's problem is given by:

$$V = \max_{c,l,e,n'} \{u(c,l) + \Phi(n')V'\}$$

subject to:

$$\begin{aligned} n(c + n'e) &= (1 - \tau)w(1 - n'\delta - l)qn \\ q' &= \Psi(e) \end{aligned}$$

Here, variables with a prime ($'$) represent those corresponding to their descendants. V and V' denote the value functions for parents and children, respectively, and $u(c,l)$ represents the instantaneous utility from consumption and leisure.

I now define a stationary equilibrium in the decentralized economy.

Definition 1 (Stationary Equilibrium). *A stationary equilibrium consists of the allocation (c, l, e, n') , aggregate quantities (Y, N) , and prices (w, τ) such that:*

- *The allocation is the solution for the household problem.*
- *Markets clear:*
 - $N = (1 - \delta n' - l)qn$
 - $Y = n(c + n'e)$
- *Given N , the wage rate is consistent with the firm's optimization behavior, that is, $w = Z$.*
- *The government budget constraint is satisfied: $G = \tau wN$*
- *Each variable is constant over time, that is, $n = n'$ and $q = q'$.*

2.3 Planner's Problem

Next, I define the planner's problem. The objective function is the same as in the decentralized economy, but the planner now internalizes the externalities arising from fertility decisions by solving a utility maximization problem subject to aggregate resource constraints. The problem is given by:

$$V = \max_{c, l, e, n'} \{u(c, l) + \Phi(n')V'\}$$

subject to:

$$\begin{aligned} Y &\equiv n^\beta AN = n(c + n'e) + G \\ N &= (1 - n'\delta - l)\Psi(e)n \end{aligned}$$

Note that the aggregate resource constraints in the planner's problem can be obtained by substituting the market-clearing conditions and the government's budget constraint into the household budget constraint of the decentralized economy. The key difference between the decentralized economy and the planner's problem is that, in the decentralized setting, households take the after-tax wage rate, $W \equiv (1 - \tau)w$, which determines their choice set, as given when making fertility decisions. In contrast, the planner fully internalizes the externalities of fertility through their effects on the equilibrium wage and tax rates.

To illustrate this, I examine how household fertility choices affect aggregate prices by differentiating the equilibrium wage and tax rates with respect to fertility:²

$$\frac{\partial W}{\partial n} = (1 - \tau) \underbrace{\frac{\partial w}{\partial n}}_{\text{Productivity}} - w \underbrace{\frac{\partial \tau}{\partial n}}_{\text{Fiscal}}$$

The first term captures the effects of fertility on technological progress and wages, which I refer to as the *productivity externality*. The second term reflects the impact of fertility on the tax rate, which I refer to as the *fiscal externality*. The productivity externality is positive because the derivative of the equilibrium wage with respect to fertility is positive:

$$\frac{\partial w}{\partial n} = \beta n^{\beta-1} A > 0 \quad (\text{Productivity externality})$$

Whether the fiscal externality is positive or negative is not clear-cut, as the effect of fertility on the equilibrium tax rate operates through four offsetting channels:

²Throughout the following analysis, I focus on a stationary equilibrium, and therefore there is no distinction between n' and n .

$$\begin{aligned}
\frac{\partial \tau}{\partial n} &= \frac{\partial \tau}{\partial Y} \left[\underbrace{\frac{\partial Y}{\partial Z} \frac{\partial Z}{\partial n}}_{\text{Technology}} + \underbrace{\frac{\partial Y}{\partial N} \frac{\partial N}{\partial n}}_{\text{Labor input}} \right] \quad (\text{Fiscal externality}) \\
&= \underbrace{-\tilde{N}\beta n^{\beta-1}A}_{\text{Technology}(<0)} \underbrace{-\tilde{Z}hq}_{\text{Increasing labor force}(<0)} \underbrace{+\tilde{Z}\delta qn}_{\text{Child penalty}(>0)} \underbrace{+\tilde{Z}hq'(n)n}_{\text{Quantity-quality trade-off}(>0)},
\end{aligned}$$

where $\tilde{N} = GN/Y^2$, $\tilde{Z} = GZ/Y^2$, $h = (1 - \delta n - l)$, and $q'(n)$ represents the optimal response of quality investment to changes in the number of children. First, higher fertility increases the stock of ideas and thereby labor productivity, which reduces the equilibrium tax rate. Second, higher fertility expands the labor force and thus enlarges the tax base, further reducing the tax rate. Third, higher fertility lowers hours worked due to the child penalty, shrinking the tax base and raising the tax rate. Fourth, higher fertility generates a quantity–quality trade-off, reducing workers’ productivity and increasing the tax rate. It is therefore unclear whether the overall fiscal externality is positive or negative. In any case, households do not internalize these externalities, which can make equilibrium fertility either too high or too low relative to the planner’s solution, in which such externalities are fully internalized.

3 Calibration

I parametrize the model using data from nine high-income countries: Canada, France, Germany, Italy, Japan, South Korea, Spain, UK, and US. Parameters are divided into country-specific parameters—preference on leisure and the quantity and quality of children, child penalties technology levels A_i , and the government expenditures G_i —and common ones—such as spillover parameter β —.

For the utility function, discount function, and human capital production function, I assume the following functional forms:

$$\begin{aligned}
u(c, l) &= \frac{(c^{1-\omega_i} l^{\omega_i})^{1-\sigma}}{1-\sigma} \\
\Phi(n') &= 1 - \exp(-\phi_i n') \\
\Psi(e) &= 1 - \exp(-\psi_i e)
\end{aligned}$$

I internally determine the preference parameters to match the fertility rate, educational attainment, and time spent in leisure in each country. Technological parameters A_i is determined to match the cross-country differences in GDP per capita. Child penalty for each country is taken from the estimate of [Kleven et al. \(2024\)](#) regarding child penalties on maternal employment. The curvature parameter σ , which governs the elasticity of intergenerational substitution (EGS), is set to 0.5 following the literature ([Cordoba et al., 2019](#)). Fig 1 shows the summary of key moments that are replicated in the baseline model across countries, and Table 1 summarizes the parameters, description, and how to determine them.

Parameter	Description	Target, reference, value
<i>Country-specific parameters</i>		
ϕ_i	Weight on child's quantity	Internally to match TFR
ψ_i	Weight on child's quality	Internally to match schooling
ω_i	Weight on leisure	Internally to match leisure share
A_i	Productivity	Internally to match GDP per worker
G_i	Social security expenditures	Internally to match expenditure-to-GDP
δ_i	Child-penalty	Externally (Kleven et al. (2024))
<i>Common parameters</i>		
σ	Curvature (EGS)	0.50 (Literature)
β	Spillover	0.35 (Externally estimated with cross-country data)

Table 1: Summary of model parameters

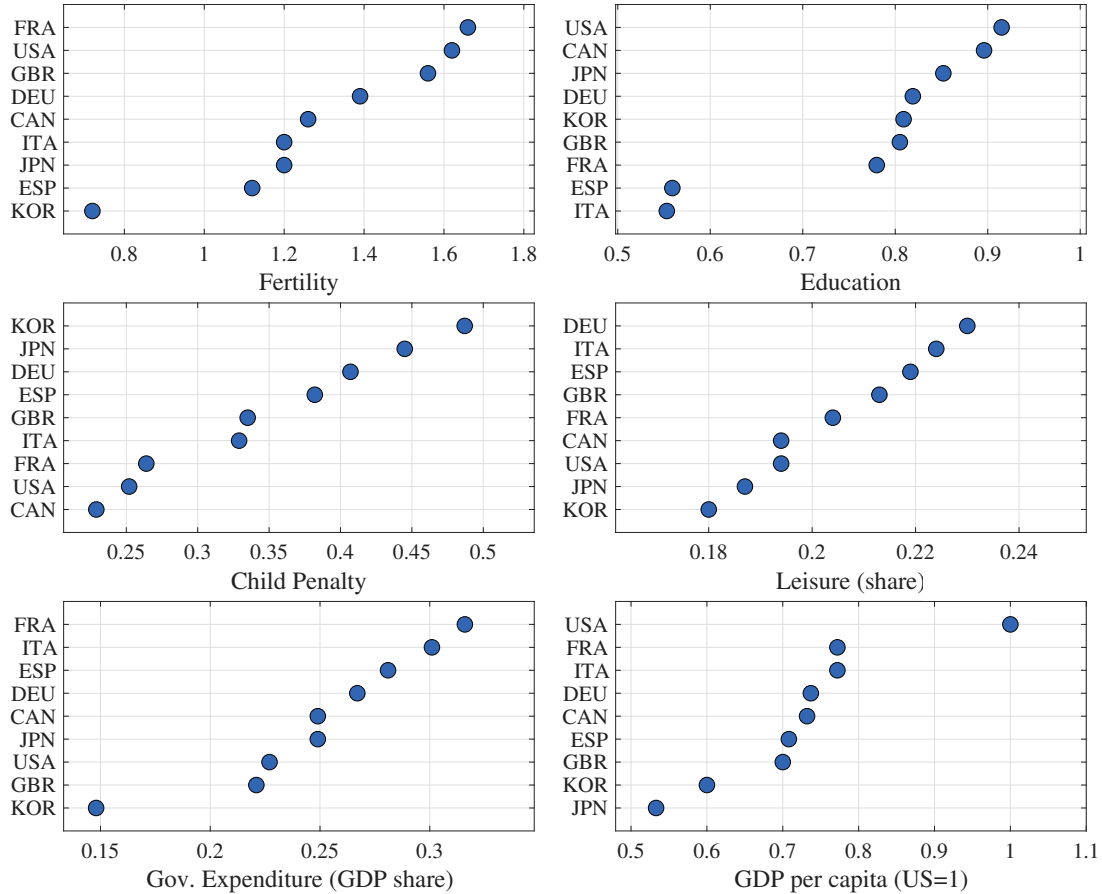


Fig 1: Targeted moments (data). *Note:* “Fertility” refers to the total fertility rates in 2023. “Education” the share of the population aged 25 or older who have completed at least upper secondary education. “Leisure” refers to the share of time allocated to leisure. “Child Penalty” refers to the employment penalty reported in [Kleven et al. \(2024\)](#), “Gov. Expenditure” refers to the share of GDP devoted to public spending on old-age security, health, unemployment, and family benefits.

To set the spillover parameter β , I estimate the effect of working-age population size on productivity using cross-country panel data from EU KLEMS (1995–2023).

The basic idea of the estimation is as follows. Recall that the aggregate production

function is given by $Y = ZN$, where $Z = n^\beta A$. Let $y \equiv Y/N$ denote output per efficiency unit of labor. Taking logs of output per efficiency unit of labor yields:

$$\ln(y) = \beta \ln(n) + \ln(A)$$

Here, β , the spillover effect of the population on technology, is the coefficient of interest. To estimate β , I consider more general production function with physical capital, to control for the effect of capital accumulation on output growth. Consider that the aggregate output in country i in year t , Y_{it} , is given by

$$Y_{it} = Z_{it} K_{it}^\alpha H_{it}^{1-\alpha},$$

where α is the capital share and H_{it} is the number of workers. Let y_{it} and k_{it} be the output and capital per worker. As in my model, consider that Z_{it} is given by $(n_{it})^\beta A_{it}$. Taking the log of both sides of the production function in terms of per worker and rearranging terms leads:

$$\underbrace{\ln(y_{it}) - \alpha \ln(k_{it})}_{\equiv e_{it}} = \beta \ln(n_{it}) + \ln(A_{it})$$

Here, n_{it} is the working-age population in country i in year t , and A_{it} explains the TFP growth unexplained by the term $\beta \ln(n_{it})$ and can contain any factors, including educational attainment, working hours per worker, R&D investments, and institutional factors. Then I consider the following regression:

$$e_{it} = \beta \ln(N_{it}) + \gamma' X_{it} + \nu_i + \eta_t + \varepsilon_{it},$$

where X_{it} contains controls such as the share of population who completed at least secondary education, the GDP share of R&D, and the average hours worked per worker. ν_i and η_t are country and time fixed effects. The coefficient on $\ln(n_{it})$ provides an estimate of the spillover effect β , where the baseline estimate results in 0.35 and is statistically significant.

4 Optimal Fertility Across Nations

Using the quantitative model, I solve the planner's problem for each country, formulated in Section 2.3. Figure 2 represents the results.

First, across all nine countries, optimal fertility exceeds its decentralized counterpart. Second, however, optimal fertility remains below the replacement rate in eight of the nine countries—with the US as the only exception. Moreover, the *fertility wedge*—defined as gaps between the planner's solution and the decentralized outcome—varies substantially across countries. For instance, Canada and Korea exhibit relatively large fertility wedge, while Germany, Japan, and Spain show relatively smaller wedges. Accordingly, the implied optimal fertility rates in these countries remain far below replacement levels.

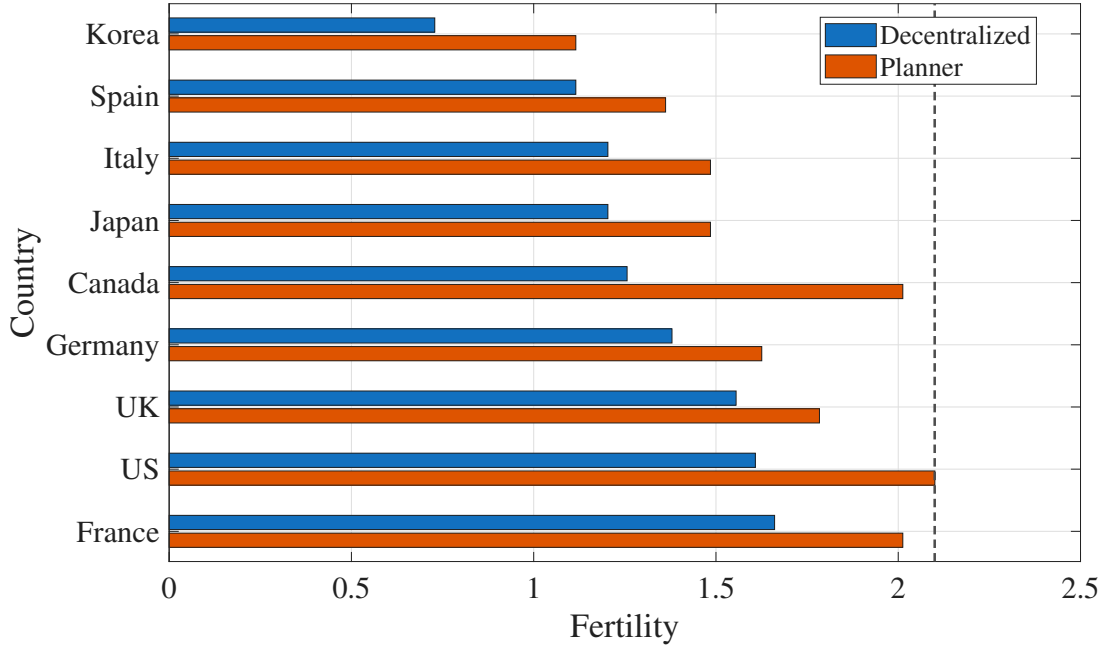


Fig 2: Fertility rates in the decentralized economy and the planner’s solution. *Note:* Fertility in the “Decentralized” economy is calibrated to match each country’s total fertility rate in 2023. The vertical dashed line at 2.10 denotes the replacement-level fertility rate.

To understand the mechanisms behind cross-country variation in the fertility wedge, I examine how it correlates with country-specific parameters. Three patterns stand out. First, child penalties are negatively associated with the fertility wedge: higher child penalties increase the social costs of fertility through reduced labor supply and output, limiting the planner’s incentive to raise fertility. This is one of the reasons why the fertility wedge in countries with high child penalties, such as Germany, Japan, and Spain, is relatively low. Second, a stronger preference for leisure is negatively associated with the fertility wedge, reducing the planner’s valuation of additional children. This is one of the reasons why the fertility wedge is relatively high in countries where people spend fewer hours on leisure, such as Canada and Korea (see bottom-right panel of Fig 1). Third, TFP is positively associated with the fertility wedge: because childrearing requires both time and financial resources, higher productivity alleviates the trade-off between having children and other uses of time and income. These results suggest that the justification for pro-natalist policy is substantially weaker in the absence of measures that reduce the social costs of higher fertility, such as mitigating the child penalty or improving labor efficiency and productivity.

5 Robustness

A key result from the baseline analysis is that, although optimal fertility exceeds the decentralized equilibrium rate, it remains below the replacement level in most countries. In this section, I examine the robustness of this result by extending the baseline model to incorporate additional mechanisms, including physical capital accumulation and the use of non-renewable resources. The main conclusion—that optimal fertility remains below the replacement level—proves to be robust across these alternative specifications. The

details of these robustness checks are discussed below.

5.1 Pollution and Climate Change

I consider the negative externalities of fertility through pollution and climate change. Specifically, I adopt a simplified version of [Gerlagh et al. \(2023\)](#), in which emissions rise with population, and the resulting climate damage drives a wedge between gross and net output. More precisely, output is produced according to the following function:

$$Y = (1 - d)ZN,$$

where the damage term d is a function of population size n and is given by

$$d = 1 - \exp(-\tilde{\delta}n),$$

where $\tilde{\delta}$ is a parameter that governs the responsiveness of climate change to an increase in population n .

I recalibrate the model with different values of $\tilde{\delta}$, ranging from 0.1 to 10, and find that the main results remain robust: incorporating this additional negative externality reduces optimal fertility.

5.2 Physical Capital

I introduce physical capital by assuming that the aggregate production function takes the form $Y = ZK^\alpha N^{1-\alpha}$ where α is the capital share and that the household problem and budget constraint are given as follows:

$$V = \max_{c, l, a', e, n'} \{u(c, l) + \Phi(n')V'\}$$

subject to:

$$c + n'(e + a') = (1 + r)a + (1 - \tau)(1 - n'\delta_i)(1 - l)wq$$

In this setup, households choose how much asset, a' , to transfer to their children. In equilibrium, the capital market clears: $K = a'$. And the interest rate r is determined by the marginal product of capital.

I set $\alpha = 0.33$, recalibrate the model, and solve the planner's problem again. The main findings remain: optimal fertility is higher than the equilibrium rate yet below replacement level in most countries, as incorporating physical capital strengthens its negative impact on capital per capita.

5.3 Fixed Resource

Building on the extension that introduces physical capital, I now consider the case in which one of the resources used in production is fixed in supply. When resources are limited, a larger population mechanically reduces the per capita allocation of the fixed resource. Following [Córdoba and Liu \(2022\)](#), I specify the production function as $Y = ZK^\alpha N^{1-\alpha}$

where the supply of K (e.g., land) is fixed at $\bar{K}$. The fixed-resource condition is therefore given by: $\bar{K} = na'$.

For each country, I set $\bar{K}$ equal to the baseline capital level computed in the recalibration with physical capital and then solve the planner's problem. The main result remains unchanged: incorporating a fixed resource amplifies the negative effect of capital dilution on capital and income per capita.

6 Concluding Remark

This paper shows that the replacement fertility rate is not a universally optimal benchmark. While optimal fertility exceeds the equilibrium rate in most countries, it remains well below replacement levels in many advanced economies, primarily due to the substantial social costs associated with higher fertility. These results suggest that the justification for pro-natalist policy is substantially weaker in the absence of measures that reduce the social costs of higher fertility, such as mitigating the child penalty or improving labor efficiency and productivity.

The current analysis treats each country as a closed economy and is therefore silent on what the optimal *global* fertility rate would be. Extending the framework to a multi-country, open-economy setting would allow for a richer examination of optimal fertility across nations. In reality, many resources—such as non-renewable resources and ideas—are at least partially shared internationally rather than being owned by a single country. These cross-country spillovers may create a wedge between the globally optimal fertility rate and the outcomes arising from decentralized economies. Incorporating such international linkages constitutes an important direction for future research.

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